

A PROJECT ON

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# Design of a High Gain Amplifier Using CMOS Technology

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**Project Stage-2 (EC 881)**

Submitted by:

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Under the Guidance of

**Dr. Anup Gorai**



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# CANDIDATE'S DECLARATION

We hereby certify that the work is being presented in the Project Report entitled **“Design of a High Gain Amplifier Using CMOS Technology”** submitted to the Department of Electronics and Communication Engineering of Ramkrishna Mahato Government Engineering College, Purulia is an authentic record of our work carried out under the supervision of Dr Anup Gorai, Asst. Professor, Department of Electronics and Communication Engineering. The matter presented in this dissertation has not been submitted by us for the award of any other degree of this or any other Institute/University.

This is to certify that the above statement made by the candidates is correct to the best of our knowledge.

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# DECLARATION BY SUPERVISOR

This is to certify that the below mentioned students has carried out the project titled “**Design of a High Gain Amplifier Using CMOS Technology**” under my guidance and supervision. Their approach has been dedicated, analytical and research-oriented.

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This project work has not been submitted elsewhere for any examination or degree.

This project is recommended for the Bachelor of Technology in Electronics and Communication Engineering at Ramkrishna Mahato Government Engineering College, Purulia affiliated to Maulana Abul Kalam Azad University, West Bengal.

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## Abstract

The increasing integration density of nanoscale CMOS technologies has created significant challenges in the development of high-gain and energy-efficient analog circuits for modern VLSI applications. Operational amplifiers are essential components in analog and mixed-signal systems used in biomedical instrumentation, communication devices, and signal processing circuits. However, deep submicron CMOS technologies suffer from reduced intrinsic gain, low output resistance, leakage current, and stability degradation, which limit the performance of conventional amplifier architectures. In this work, a high-gain and low-power CMOS operational amplifier is designed and analyzed using 90 nm CMOS technology. The proposed amplifier employs optimized transistor sizing, current mirror biasing, and gain enhancement techniques to achieve improved voltage gain, enhanced bandwidth, and stable operation while maintaining low power dissipation and compact circuit architecture. The complete circuit design and simulation were carried out using Cadence Virtuoso environment. Extensive circuit-level simulations were performed to evaluate important performance parameters, including open-loop gain, phase margin, bandwidth, and power consumption. The simulation results demonstrate that the proposed amplifier achieves a high open-loop gain of 66.5893 dB with a phase margin of 71.8°, indicating stable and reliable operation. In addition, the amplifier exhibits a bandwidth of 25.233 MHz and consumes only 290.462  $\mu$ W of power, making it suitable for low-power and high-speed analog applications. Compared with conventional CMOS operational amplifier designs, the proposed architecture provides improved gain, enhanced stability, and better frequency response without increasing circuit complexity, making it suitable for modern analog VLSI and mixed-signal integrated circuit applications.

To further extend the proposed design, the operational amplifier was implemented as a three-op-amp CMOS instrumentation amplifier. The resulting instrumentation amplifier achieved a significantly improved open-loop gain of 106.083 dB and a Common-Mode Rejection Ratio (CMRR) of 110.223 dB, while consuming only 1 mW of power. These results demonstrate excellent differential amplification capability, superior common-mode noise rejection, and high precision, making the design well suited for biomedical signal acquisition, sensor interfacing, and other precision analog applications. Compared with conventional CMOS amplifier architectures, the proposed designs provide enhanced gain, improved stability, superior frequency response, and excellent common-mode rejection without significantly increasing circuit complexity, making them attractive candidates for next-generation analog VLSI and mixed-signal integrated circuit applications.

# **Chapter 1**

## **Introduction**

Operational amplifiers (Op-Amps) are key components in modern analog and mixed-signal integrated circuits. They are widely used in data converters, sensor interfaces, communication systems, biomedical instrumentation, active filters, voltage regulators, and various signal-conditioning applications. The ongoing advancement of CMOS technology has allowed for the creation of compact, low-power, and high-speed integrated circuits, but it also poses significant challenges for analog circuit design. As transistor sizes shrink into the deep-submicron range, the inherent gain of transistors drops because of lower output resistance, reduced supply voltages, greater channel-length modulation, and marked short-channel effects. This makes it tough to achieve high open-loop gain while ensuring enough bandwidth, stability, and low power use. This challenge is central to modern CMOS operational amplifier design [1]-[3].

Among the many operational amplifier designs, the two-stage CMOS operational amplifier is one of the most popular. It strikes a good balance between voltage gain, output swing, driving capability, and implementation complexity. Usually, the first stage includes a differential input amplifier with active current-mirror loads that provide high input impedance and differential amplification. The second stage uses a common-source gain stage to boost voltage gain and enhance output driving capability [1]. Proper frequency compensation, typically done with Miller compensation, ensures a good phase margin and stable closed-loop operation even with multiple poles [2].

As CMOS technology continues to advance, analog designers have proposed several methods to enhance gain and tackle the drop in intrinsic device gain. These methods include multi-stage amplifier designs, compound transconductance structures, improved current mirrors, active loads, gain boosting, and positive feedback techniques [2], [3]. Such methods can achieve open-loop gains over 100 dB while keeping gain-bandwidth product and stability at acceptable levels. This makes them ideal for high-resolution analog signal processing systems. However, these techniques often involve trade-offs among power consumption, silicon area, compensation complexity, and design robustness.

Instrumentation amplifiers (INAs) are another vital type of analog circuit that accurately amplify very small differential signals, even in high common-mode voltage situations. With their very high input impedance, high common-mode rejection ratio (CMRR), low offset voltage, and programmable gain, instrumentation amplifiers are commonly used in biomedical signal acquisition, industrial measurement systems, sensor interfaces, strain gauges, thermocouples, and data acquisition systems [4], [5]. In applications like electrocardiogram (ECG), electroencephalogram (EEG), and other physiological monitoring systems, input signals usually fall within the microvolt or millivolt range and are very sensitive to common-mode interference and environmental noise. Therefore, instrumentation amplifiers must deliver high differential gain, excellent CMRR, low noise, and low power consumption [4], [5].

The traditional three-op-amp instrumentation amplifier design remains the favourite choice due to its high input impedance, excellent gain accuracy, and outstanding common-mode rejection. The performance of this design heavily relies on the qualities of the operational amplifiers used. Thus, creating a high-gain, low-power operational amplifier with enough bandwidth and stability directly boosts the overall performance of the instrumentation amplifier. [4], [5]

Recent research has shown various CMOS operational amplifier designs in technology nodes from 180 nm to 22 nm to tackle the issues that come with technology scaling. Multi-stage amplifier setups using compound transconductance elements have achieved open-loop gains over 100 dB while maintaining good gain-bandwidth features [2]. Additionally, improved current-mirror structures, active-load techniques, and optimized transistor sizing have significantly increased voltage gain without excessive power use [3], [5]. Recent instrumentation amplifier designs in 90 nm CMOS technology have shown that replacing passive resistive loads with active NMOS loads can

reduce chip area and power consumption while also improving differential gain and common-mode rejection ratio [5].

Inspired by these advancements, this work presents the design and simulation of a high-gain two-stage CMOS operational amplifier built with 90 nm CMOS technology in the Cadence Virtuoso design environment. This operational amplifier serves as the basic component of a conventional three-op-amp instrumentation amplifier. The goal is to achieve high open-loop gain, adequate bandwidth, good phase margin, low power consumption, and excellent common-mode rejection suitable for precision analog signal conditioning and sensor interface applications. The performance of both the operational amplifier and the instrumentation amplifier is assessed through thorough simulations to validate their effectiveness for modern low-power analog integrated circuit applications.

## **Chapter 2**

### **Overview of the Proposed Amplifier**

Operational amplifiers (Op-Amps) are crucial components in modern integrated circuits. They carry out a variety of tasks, such as amplifying signals, filtering, buffering, performing mathematical operations, converting data, and regulating voltage. Because of their versatility and high performance, operational amplifiers are commonly used in communication systems, biomedical devices, sensor interfaces, industrial control systems, and mixed-signal integrated circuits [1]-[4].

The ongoing progress of CMOS technology has led to the creation of compact, fast, and low-power integrated circuits. As technology moves into the deep-submicron range, like the 90 nm CMOS process, engineers achieve higher transistor density and better switching speeds. Nonetheless, scaling down technology brings several analog design problems, including reduced intrinsic transistor gain, lower supply voltages, higher leakage currents, and short-channel effects. These issues make it harder to achieve high open-loop gain while also ensuring wide bandwidth, adequate stability, and low power consumption [2], [3], [5].

To overcome these challenges, various CMOS operational amplifier designs have been proposed, such as folded-cascode, telescopic, multistage, and two-stage topologies. Among these, the two-stage CMOS operational amplifier remains one of the most commonly used designs. It strikes a good balance between voltage gain, output swing, bandwidth, circuit complexity, and power efficiency. Its relatively straightforward design also makes it suitable for precision analog signal-conditioning circuits in modern CMOS technologies [2], [5], [6].

### **3.1 Two-Stage CMOS Operational Amplifier**

The suggested op amp is built in the traditional two-stage structure with 7 MOS transistors and one compensation capacitor. The first stage consists of M1 and M2 which form the differential input pairs and receive  $V_{in}^+$  and  $V_{in}^-$ , the input signals to be amplified. M1 and M2 can provide high input impedance.

As the base amplifier structure, differential pair amplification amplifies only the difference voltage of the two-input terminal and reject the common-mode signal and improves the noise immunity [1],[4].

Transistors M3 and M4 act as active current-mirror load for the differential pairs. Unlike the use of passive resistors as the loads for the differential input pairs, active loads give much higher output impedance and can greatly reduce the occupied chip area. This also results in a high voltage gain for the first stage without adding to the total power consumption [2],[5]. The current needed by the differential input pair is supplied by transistor M5, the constant current source.

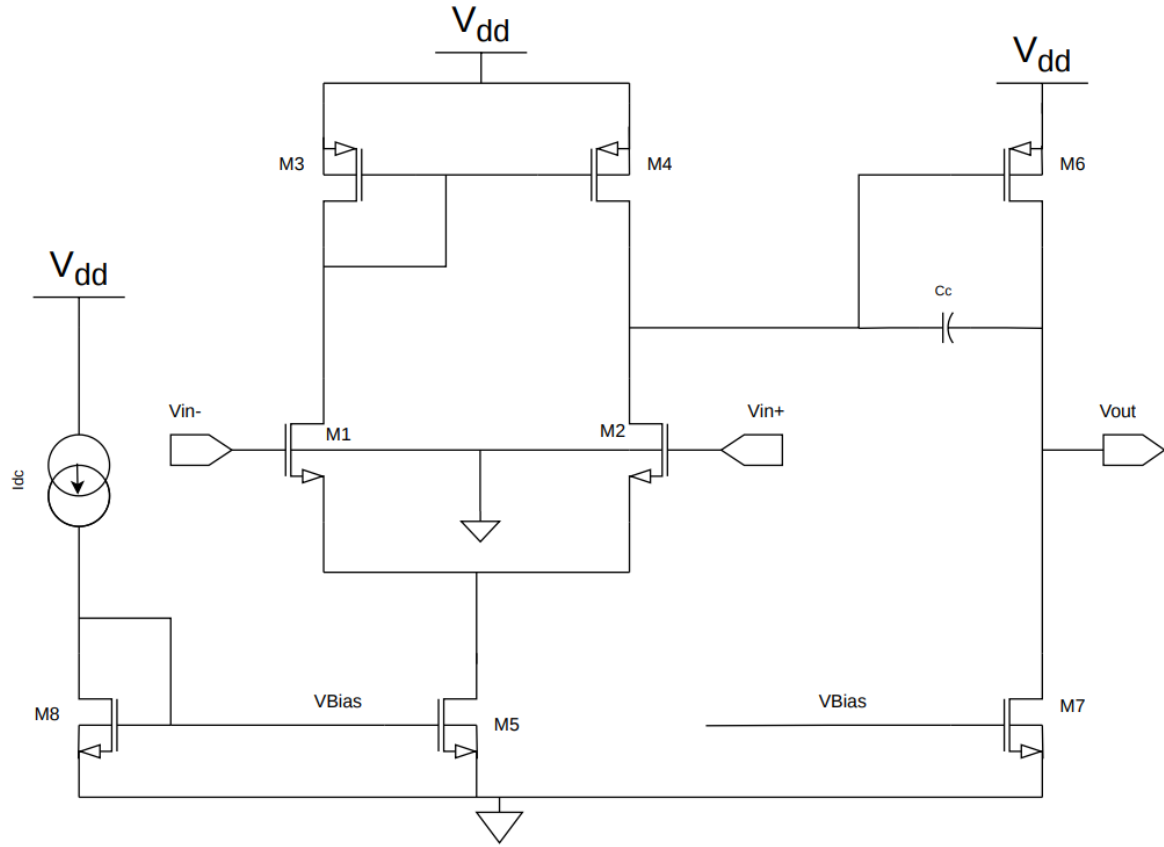
The tail current is determined by transistor M8 which forms the biasing of the amplifier.

Bias ensures the stability and linearity of the amplifier. By providing adequate tail current and voltage swing, transistors can operate in the saturation region [2],[3]. Transistor M6 is the common-source amplification in the second stage.

This stage provides additional voltage gain and improves the driving capacity of the op amp. Transistor M7 provides active load for the second stage. And this makes the amplifier deliver higher output voltage swing and more gain [2].

In order to ensure stability under negative feedback, the Miller compensation capacitor  $C_c$  is inserted between the output of the second stage and the output of the first stage.

This improves the phase margin by making a dominant pole in the frequency response, and so avoids oscillation [2],[3]. We choose this architecture because of its advantages of high open loop gain, bandwidth, phase margin and low power consumption, and the simplicity for designing with 90nm CMOS technology.



**Fig. 1.** A conventional two-stage operational amplifier

### 3.2 Instrumentation Amplifier

An instrumentation amplifier is a high precision analog signal conditioning amplifier, which takes two very high input impedance single ended differential amplifiers in an integrated circuit instrumentation, amplifies very low differential input signal with respect to common mode, rejects unwanted common mode and noise voltage and has highly precision and stable gain [4],[6],[7].

Figure 2. Most widely used and versatile instrumentation amplifier topology, it provides high accuracy for the differential gain, and high input impedance, high common mode rejection. There are two buffer front end op-amps, where one of the inputs of both amplifiers are directly fed from either  $V_1$  or  $V_2$  which would provide a buffer from the input and the first gain stage. The second gain stage of the circuit is provided by the following stage which is the op amp and the combination of the resistors  $R_1$ - $R_{gain}$ - $R_1$  which is connected between the outputs of the first two buffers and acts to provide the first gain for the differential signal and to enable common mode signals to pass to the third amplifier without gain as the current flowing through  $R_{gain}$  is zero under common mode operation [4],[6].

For the topology in Figure 2, the overall output voltage is given by:

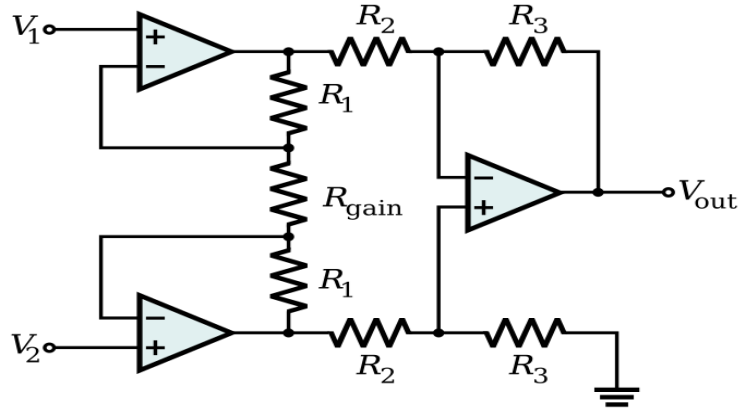
$$V_{out} = \left( \frac{2R_1}{R_{gain}} \right) \frac{R_3}{R_2} (V_2)$$

In this configuration the first stage introduces a differential gain of  $\left(\frac{2R_1}{R_{gain}}\right)$ , this differential gain can be tuned by one resistor  $R_{gain}$  without upsetting the symmetry of the front-end. The second stage adds a fixed gain of  $\frac{R_3}{R_2}$ , set by the difference amplifier resistor matching. By separating the common-mode rejection and gain setting elements into different stages, the 3 op-amp instrumentation amplifier topology gains a higher CMRR over its single op-amp counterpart, the overall performance of the instrumentation amplifier depends critically on the internal op-amp's open loop gain, offset voltage and bandwidth etc [4],[6].

In this paper, the proposed high-gain two-stage CMOS op-amp has been used to design a three-op-amp instrumentation amplifier. The full design and simulation are performed in Cadence Virtuoso IC Design tool using GDPK90 CMOS technology Library. The designed instrumentation amplifier will be characterized by differential gain, common-mode rejection, frequency response and power consumption appropriate to analog signal-conditioning application.

Recently high performance, small size, low power, high gain, high CMRR and good enough frequency response of CMOS Instrumentation Amplifier and acceptable performance on VLSI are the subjects of intense research activities [5],[6].

The instrumentation amplifier, especially in advanced CMOS technology, such as 90 nm CMOS process technology, are the favored approach for application on the low-power VLSI systems because of device size reduction, integration capability improvement and performance enhanced.



**Fig. 2.** Conventional operational amplifier-based INA

## Chapter 3

### Circuit Diagram and Methodology

#### 4.1 Design of the Proposed Two-Stage CMOS Operational Amplifier

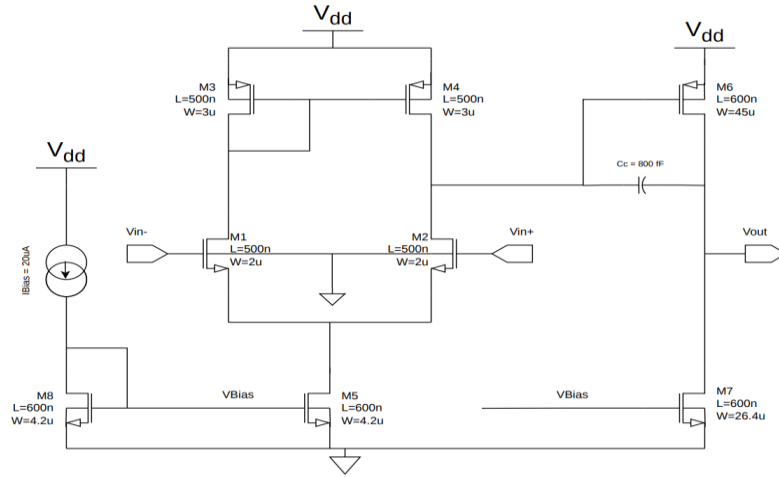
This amplifier is laid out in schematic capture & simulating using Cadence Virtuoso Analog design Environment with Generic Process design Kit (GPD90) CMOS Technology. The design objective was to achieve high open-loop gain, adequate phase margin, wide bandwidth, and low power consumption suitable for precision analog applications. The target specifications were selected based on recent CMOS operational amplifier designs reported in the literature [1] – [5].

The proposed operational amplifier employs a conventional two-stage CMOS topology, as shown in **Fig. 3**.

The circuit consists of:

- Differential input pair (**M1–M2**)
- PMOS current mirror active load (**M3–M4**)
- NMOS tail current source (**M5**)
- PMOS common-source second stage (**M6**)
- NMOS active load (**M7**)
- Bias current transistor (**M8**)
- Miller compensation capacitor (**C<sub>c</sub>**)

The differential input stage converts the differential input signal into a single-ended output while providing high input impedance and initial voltage gain. The second stage provides additional voltage gain and improves output driving capability. Miller compensation is employed to ensure closed-loop stability [2], [3].



**Fig. 3.** Proposed Schematic of two-stage CMOS operational amplifier

The bias current is obtained from the required slew rate.

$$I_{Bias} = SR \times C_c$$

Where:

- SR = Slew Rate
- C<sub>c</sub> = Compensation Capacitor

The compensation capacitor/Miller capacitor is selected from the load capacitance. The commonly used approximation is:

$$C_c \geq 0.22 \times C_L \quad [C_L = \text{Load capacitance}]$$

The required transconductance of the input pair is

$$g_m = 2\pi \text{GBW} \times C_c$$

This equation determines the required input transistor size.

The overdrive voltage  $V_{OV} = V_{GS} - V_{TH}$

Drain current  $I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{ov})^2$

From this equation we can get the equation of the aspect ratio of the CMOS

$$\frac{W}{L} = \frac{2I_D}{\mu C_{ox} (V_{ov})^2}$$

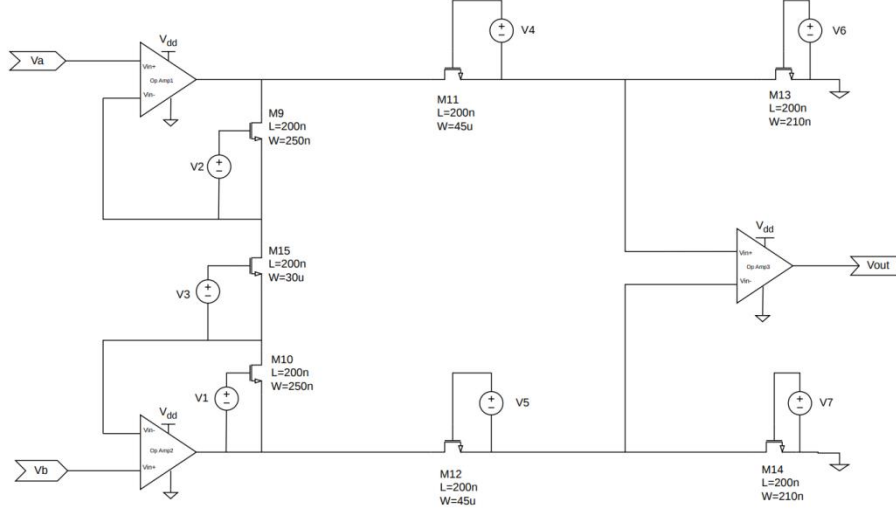
Using This equation, we calculated the aspects ratio of the CMOS as shown in table 1.

**Table 1.** Dimensions of the CMOS of the proposed two-stage operational amplifier

| CMOS Number | Length | Width |
|-------------|--------|-------|
| M1, M2      | 500n   | 2μ    |
| M3, M4      | 500n   | 3μ    |
| M5, M8      | 600n   | 4.2μ  |
| M6          | 600n   | 45μ   |
| M7          | 600n   | 26.4μ |

## 4.2 Design of the Proposed Instrumentation Amplifier

The suggested instrumentation amplifier circuit has been designed with three identical two-stage CMOS operational amplifiers designed in the previous section. The entire circuit has been built using the Cadence Virtuoso Analog Design Environment tool with Cadence GPD90 CMOS technology. The instrumentation amplifier does not use the conventional precise resistor networks to set the gain. Rather the network is built from the MOS transistors to determine the flow of currents between the two amplification stages, thus resulting in a more adaptable architecture for integrated CMOS realization where gain is defined through the transistors dimensions instead of resistors ratio.



**Fig. 4.** Proposed Schematic of Instrumentation amplifier

The first stage consists of two identical two-stage CMOS operational amplifiers connected in a symmetrical configuration. The input voltages  $V_a$  and  $V_b$  are applied to the non-inverting terminals of the operational amplifiers, while the inverting terminals are interconnected through the MOS transistor network consisting of **M9**, **M10**, and **M15**.

This stage provides:

- High input impedance
- Initial differential amplification
- Balanced signal processing
- Improved common-mode rejection

Since both operational amplifiers are identical, equal gain and bandwidth are maintained for the two input channels.

The first stage consists of **two identical two-stage operational amplifiers**. Each amplifier has an open-loop gain

$$A_{OP} = A_{v1} \times A_{v2}$$

where

$$A_{v1} = g_{m1}(r_{o4})$$

is the gain of the differential stage, and

$$A_{v2} = g_{m6}(r_{o7})$$

is the gain of the second common-source stage.

Because the two input amplifiers are identical,

$$A_{OP1} = A_{OP2} = A_{OP}$$

The MOS current mirror formed by the transistors scales the signal according to the aspect ratio

$$A_{CM1} = \frac{\left(\frac{W}{L}\right)_{M15}}{\left(\frac{W}{L}\right)_{M9}}$$

Since all devices have

$$L = 200 \text{ nm},$$

The previous equation becomes

$$A_{CM1} = \frac{W_{M15}}{W_{M9}} \text{ Similarly, } A_{CM2} = \frac{W_{M15}}{W_{M10}}$$

Hence, the effective gain of the first stage can be approximated as

$$A_{Stage1} = A_{OP} \left(\frac{W_{M15}}{W_{M9}}\right) \text{ or } A_{Stage1} = A_{OP} \left(\frac{W_{M15}}{W_{M10}}\right) [\text{Due to symmetry}]$$

The third operational amplifier is **the same two-stage CMOS operational amplifier** used in the first stage.

Therefore,

$$A_{OP3} = A_{OP}$$

The current mirror associated with the output stage is

$$A_{CM3} = \frac{\left(\frac{W}{L}\right)_{M11}}{\left(\frac{W}{L}\right)_{M13}} \text{ and } A_{CM4} = \frac{\left(\frac{W}{L}\right)_{M12}}{\left(\frac{W}{L}\right)_{M14}}$$

Since all channel lengths are identical,

$$A_{CM3} = \frac{W_{M11}}{W_{M13}} \text{ and } A_{CM4} = \frac{W_{M12}}{W_{M14}}$$

The gain of the differential output stage is therefore

$$A_{Stage2} = A_{OP} \left(\frac{W_{M11}}{W_{M13}}\right) \text{ or } A_{Stage2} = A_{OP} \left(\frac{W_{M12}}{W_{M14}}\right)$$

**Table 2.** Dimensions of the required CMOS of the proposed instrumentation amplifier

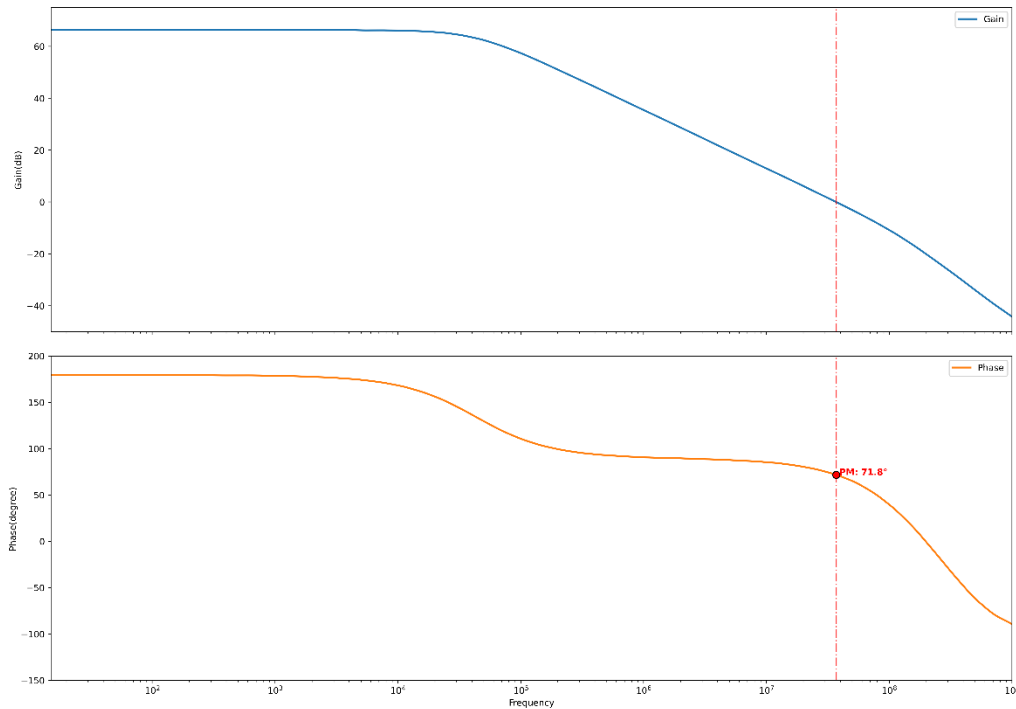
| MOS Number | Length | Width |
|------------|--------|-------|
| M9, M10    | 200n   | 250n  |
| M11, M12   | 200n   | 45μ   |
| M13, M14   | 200n   | 210μ  |
| M15        | 200n   | 30μ   |

## Chapter 4

### Simulation and Results

A two-stage CMOS operational amplifier and a two-stage CMOS instrumentation amplifier was designed using the Cadence Virtuoso Analog Design Environment with the Cadence Generic Process Design Kit (GPD90) CMOS technology library. The simulations for the proposed circuits were implemented in Cadence Virtuoso through the Analog Design Environment L (ADE L). The two-stage CMOS operational amplifier was first designed and tuned for necessary electrical characteristics and after verifying the performances of the designed operational amplifier, three copies of the operational amplifiers were connected in parallel to realize a two-stage CMOS instrumentation amplifier. The dc operating point analysis, ac frequency response analysis, and transient analysis of both the two-stage CMOS operational amplifier and the two-stage CMOS instrumentation amplifier were implemented in the ADE L.

The designed CMOS two-stage op-amp is evaluated for DC gain, phase margin, bandwidth and power dissipation. From the simulation results shown above the circuit provides stable performance with sufficient gain and wide frequency response.



**Fig. 5.** Plot of Gain and Phase vs frequency response of the proposed two stage operational amplifier

The proposed amplifier achieved a **DC gain of 66.59 dB**, demonstrating its capability to provide substantial voltage amplification suitable for precision analog applications.

A **phase margin of 71.8°** was obtained, confirming stable operation under negative feedback and indicating the effectiveness of the Miller compensation technique employed in the proposed design. A phase margin greater than 60° is generally considered sufficient for stable operation.

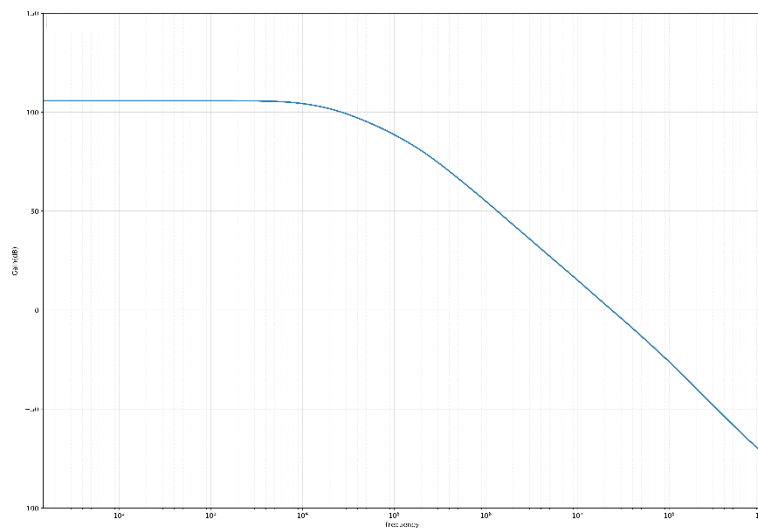
The proposed amplifier achieved a **bandwidth of 25.23 MHz**, providing a wide operating frequency range suitable for high-speed analog signal processing applications.

The total simulated **power dissipation was 290  $\mu$ W**, indicating that the proposed design maintains low power consumption while achieving high gain and satisfactory frequency response.

**Table 3.** Performance Comparison of the Proposed Two-Stage CMOS Operational Amplifier

| Parameters                          | Reference Work [8] | Referenc e Work [7] | Referenc e Work [9] | Referenc e Work [6] | Referen ce Work [10] | Referen ce Work [5] | Our Work |
|-------------------------------------|--------------------|---------------------|---------------------|---------------------|----------------------|---------------------|----------|
| Technology                          | 0.18 $\mu\text{m}$ | 0.35 $\mu\text{m}$  | 90 nm               | 90 nm               | 0.18 $\mu\text{m}$   | 90 nm               | 90 nm    |
| DC Gain (dB)                        | 76.18              | 33.7                | 49                  | 47.7                | 72.1                 | 69.39               | 66.59    |
| Phase Margin(degr ee)               | 60                 | --                  | --                  | --                  | --                   | --                  | 71.8     |
| Bandwidth (MHz)                     | 42.363             | 2                   | 0.27                | 1.25                | 0.32                 | 6.38                | 25.23    |
| Power Dissipation ( $\mu\text{W}$ ) | 855                | 850                 | --                  | 300                 | 309.45               | 391                 | 290      |

The instrumentation amplifier was implemented using three identical two-stage CMOS operational amplifiers designed in this work. The complete circuit was simulated using the Cadence GPD90 CMOS technology library under identical operating conditions.

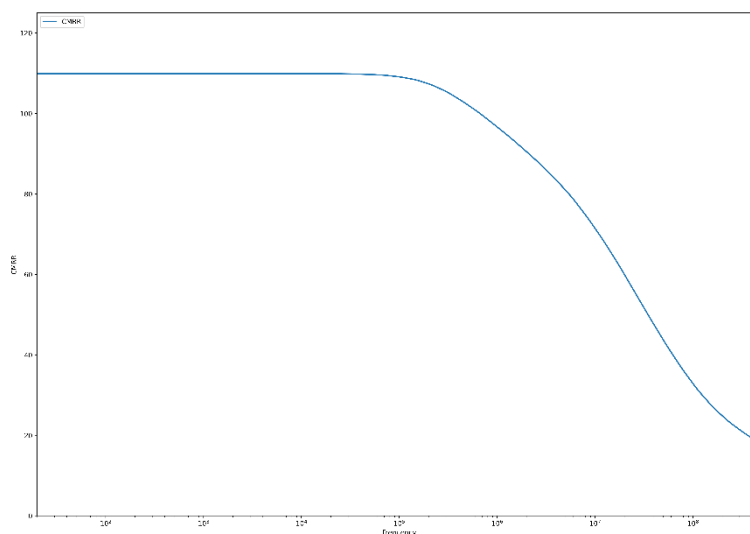
**Fig. 6.** Plot of Gain vs Frequency response of the proposed instrumentation amplifier

The proposed instrumentation amplifier achieved a **DC gain of 106.08 dB**, demonstrating excellent amplification capability for low-amplitude differential signals and precision analog signal-conditioning applications.

A **common-mode rejection ratio (CMRR) of 110.22 dB** was obtained, indicating effective suppression of common-mode interference and making the proposed design suitable for sensor interface and biomedical instrumentation applications.

The simulated **gain-bandwidth product (GBW) was 23.96 MHz**, demonstrating that the proposed instrumentation amplifier is capable of providing high gain while maintaining a wide frequency response.

The complete instrumentation amplifier consumed **1 mW** of power. The increase in power consumption is expected because the circuit employs **three identical two-stage CMOS operational amplifiers** together with the associated MOS biasing circuitry.



**Fig. 7.** Plot of Common Mode Rejection Ratio (CMRR) vs Frequency response of proposed instrumentation amplifier

**Table 4.** Performance Comparison of the Proposed Instrumentation Amplifier

| Parameters                          | Referenc<br>e Work<br>[7] | Referenc<br>e Work<br>[9] | Referenc<br>e Work<br>[6] | Referen<br>ce Work<br>[5] | Our<br>Work |
|-------------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|-------------|
| Technology                          | 0.35 $\mu\text{m}$        | 90nm                      | 90nm                      | 90nm                      | 90nm        |
| DC Gain (db)                        | 33.7                      | 49                        | 47.7                      | 97.69                     | 106.08      |
| CMRR (dB)                           | 83                        | 108                       | 98.3                      | 135.72                    | 110.22      |
| Gain Bandwidth Product (MHz)        | 2                         | 0.27                      | 1.25                      | 4.52                      | 23.96       |
| Power Dissipation ( $\mu\text{W}$ ) | 850                       | --                        | 300                       | 870                       | 1000        |

## **Chapter 5**

### **Discussion**

The results obtained from the proposed high-gain CMOS operational amplifier and instrumentation amplifier in 90 nm CMOS technology demonstrate notable advancements in gain, stability, bandwidth, and power efficiency when compared against the broader literature.

Designing high-gain analog circuits in nanoscale CMOS is inherently challenging. As technology scales into the deep-submicron regime, transistors exhibit reduced intrinsic gain, lower output resistance, increased leakage currents, and pronounced short-channel effects — collectively degrading achievable open-loop gain [1]– [3]. To address these challenges, analog designers have proposed techniques such as multi-stage amplifier topologies, compound transconductance structures, improved current mirrors, active loads, and gain boosting — methods capable of achieving open-loop gains exceeding 100 dB while maintaining acceptable gain-bandwidth products [2], [3].

Among available CMOS op-amp topologies, the two-stage architecture remains the most widely adopted due to its balanced trade-off between voltage gain, output swing, bandwidth, and implementation simplicity [1], [2]. In the proposed design, the first stage provides high input impedance and initial voltage gain through a differential pair with active current-mirror loads, while the second stage further boosts gain and driving capability via a common-source amplifier with an active NMOS load. Miller compensation via  $C_c$  establishes a dominant pole, ensuring stable closed-loop operation [2], [3]. The proposed op-amp achieves a DC gain of **66.59 dB**, a phase margin of **71.8°**, a bandwidth of **25.23 MHz**, and a power dissipation of only **290  $\mu$ W**. The phase margin of 71.8° — well above the 60° threshold generally considered sufficient for stable operation — reflects the effectiveness of the Miller compensation strategy. For comparison, [3] achieved a higher DC gain of 103 dB in a 22 nm FDSOI process using a modified high output-impedance current mirror, but at a phase margin of only 50°, indicating that the proposed design offers superior stability robustness despite operating at a less advanced technology node.

The proposed instrumentation amplifier, built from three identical two-stage CMOS op-amps in the classic three-op-amp topology, achieved a DC gain of **106.08 dB**, a CMRR of **110.22 dB**, a GBW of **23.96 MHz**, and a power dissipation of **1 mW**. The gain network is implemented using MOS transistors rather than conventional precision resistors, yielding a more area-efficient and VLSI-compatible architecture where gain is defined by transistor aspect ratios rather than resistor ratios. Compared against state-of-the-art designs, Reference [7] (0.35  $\mu$ m) reported only 33.7 dB gain and 2 MHz GBW, while [9] (90 nm) achieved 49 dB gain at a very narrow GBW of 0.27 MHz. [5] (90 nm) reported a higher CMRR of 135.72 dB, but a GBW of only 4.52 MHz — more than five times narrower than the proposed design's 23.96 MHz. The proposed design thus achieves the highest DC gain and widest GBW among all compared 90 nm designs, with the slight increase in power consumption to 1 mW being well justified by these performance gains.

## Chapter 6

### Conclusions

This work presented the design, simulation, and performance evaluation of a high-gain, low-power two-stage CMOS operational amplifier and its extension into a three-op-amp instrumentation amplifier, both implemented using 90 nm CMOS technology in the Cadence Virtuoso Analog Design Environment with the GPD90 process design kit.

The proposed two-stage CMOS operational amplifier was designed using a conventional topology comprising a differential input pair (M1–M2) with a PMOS current mirror active load (M3–M4), an NMOS tail current source (M5), a PMOS common-source second stage (M6), an NMOS active load (M7), a bias current transistor (M8), and a Miller compensation capacitor ( $C_c$ ). The design philosophy centered on achieving a balanced trade-off between voltage gain, bandwidth, phase margin, and power dissipation — parameters that are inherently difficult to optimize simultaneously in deep-submicron CMOS technologies due to reduced intrinsic device gain, short-channel effects, and lower supply voltages.

Simulation results confirmed that the proposed op-amp achieved a DC open-loop gain of **66.59 dB**, a phase margin of **71.8°**, a bandwidth of **25.23 MHz**, and a power dissipation of only **290  $\mu$ W**. The phase margin exceeding 60° validates the effectiveness of the Miller compensation technique in ensuring closed-loop stability under negative feedback. The wide bandwidth of 25.23 MHz makes the design well-suited for high-speed analog signal processing, while the low power dissipation of 290  $\mu$ W confirms its suitability for energy-constrained applications.

Building upon the verified op-amp design, a three-op-amp instrumentation amplifier was realized by interconnecting three identical copies of the proposed op-amp with a MOS transistor-based gain-setting network, replacing conventional precision resistor networks. This architecture inherently provides high input impedance, programmable differential gain, and superior common-mode rejection. The instrumentation amplifier achieved a remarkable DC gain of **106.08 dB**, a CMRR of **110.22 dB**, and a gain-bandwidth product of **23.96 MHz**, while consuming **1mW** of total power — a predictable and acceptable increase given the use of three op-amp cores along with associated biasing circuitry.

Comparative analysis with previously reported CMOS op-amp and instrumentation amplifier designs across technology nodes ranging from 0.18  $\mu$ m to 90 nm demonstrated that the proposed designs offer competitive and, in several key metrics, superior performance. Notably, the proposed op-amp delivers a higher phase margin than several reference designs, and the instrumentation amplifier achieves the highest DC gain and gain-bandwidth product among the 90 nm counterparts compared in this study.

In summary, the proposed designs successfully address the key challenges of analog circuit design in nanoscale CMOS technologies — namely, achieving high gain and wide bandwidth while maintaining stability and low power consumption. The results validate the design methodology and confirm that the proposed two-stage CMOS op-amp and instrumentation amplifier are strong candidates for deployment in precision analog applications, including biomedical signal acquisition (ECG, EEG), sensor interfacing, industrial measurement systems, and mixed-signal VLSI circuits. Future work may explore gain-boosting techniques, noise optimization, layout-level implementation with post-layout simulation, and extension to lower supply voltage operation to further enhance the applicability of the proposed designs in next-generation analog integrated circuits.

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